LESSON 3.3

Determine Equivalent Ratios Using Tables

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use a ratio table to find equivalent ratios.

WHY THIS MATTERS Brian will use the recipe shown to make clay. He needs to make multiple batches of clay. How can Brian determine how much baking soda and cornstarch he needs for multiple batches of clay to get the same consistency?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Ratio Tables *Spanish: Tablas de razones*	Tables that organize pairs of equivalent ratios.	2:3, 4:6, and 6:9 belong in the same ratio table.
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that shows a pattern.	5, 10, 15 / 2, 4, 6
Equivalent ratio *Spanish: Razón equivalente*	Two ratios that mean the same thing, like 1:2 and 2:4.	2:3 = 4:6 = 6:9
Scale factor *Spanish: Factor de escala*	The number you multiply both parts of a ratio by to get an equal ratio.	2:3 × 2 → 4:6 (scale factor is 2)
Pattern *Spanish: Patrón*	A rule that numbers follow again and again.	Each row adds 2 cups juice and 3 cups water
Additive pattern *Spanish: Patrón aditivo*	Building new rows of a ratio table by adding each column's own constant amount, so the two columns grow together.	2:3 → 4:6 → 6:9 (add 2 and 3 each time)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A mix uses 5 scoops of powder for every 2 cups of water. How much water for 15 scoops?

1. Find the factor: 15 ÷ 5 = 3.
2. Multiply the water by 3 too: 2 × 3 = 6.
3. Check: 15 : 6 ÷ 3 = 5 : 2.

Answer: 15 scoops go with 6 cups of water

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A drink uses 1 cup of juice for every 4 cups of water, so the ratio is 1:4.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I multiplied both terms by _____ .
- _____ and _____ are equivalent because _____ .
- Adding does not work because _____ .
- The factor is _____ because _____ × _____ = _____ .

Word bank equivalent | ratio table | factor | multiply | divide | both | same | scale up | scale down | for every | row | pattern

WATCH OUT!

A common mistake

A common mistake in Ratio Tables is scaling only ONE part of the ratio, or adding the scale factor to a part instead of multiplying by it

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

A recipe uses 3 eggs for every 5 cups of flour. How many eggs are needed for 15 cups of flour?

(A) 6
(B) 9
(C) 10
(D) 12

STEPS TO TRY
1. Notice the flour jumped from 5 cups to 15 cups — the eggs must grow by the same factor.
2. Find the scale factor by dividing the new flour by the recipe's flour, then multiply the eggs by that same factor.
3. Divide 15 ÷ 5 to get the scale factor. Then multiply 3 eggs by that factor to scale the recipe.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which row does NOT belong in a ratio table for 4:5?

(A) 6:10
(B) 8:10
(C) 12:15
(D) 16:20

STEPS TO TRY

1. Every row in a 4:5 table must simplify back to exactly 4:5.
2. For each choice, ask: can I multiply 4 AND 5 by the same number to get these two values? Check both parts, not just one.
3. Work through the choices in order. For each one ask: is there a single number that multiplies BOTH 4 and 5 into these two values?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A ratio table shows 3:7, 6:14, and 9:21. What is the next row?

(A) 10:24
(B) 12:28
(C) 12:24
(D) 15:28

STEPS TO TRY

1. Look down each column: how does 3 → 6 → 9 change? How does 7 → 14 → 21 change?
2. Each new row adds the original ratio once more: +3 on the left and +7 on the right. Apply that jump to the last row.
3. Find the next left value first: 9 + 3. Then figure out what must be added to 21 to keep the right column's pattern.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A ratio table for 2:5 starts with 2:5, 4:10. What is the third row?

(A) 5:10
(B) 6:15
(C) 6:12
(D) 8:15

STEPS TO TRY

1. Compare row 1 (2:5) with row 2 (4:10). What happened to each column?
2. The table grows by one batch each row: the left column climbs by 2 and the right column climbs by 5.
3. Work the left column first: add 2 to the 4. Then decide what the right column must be so the row still matches 2:5.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you find each new row in the pancake ratio table? If someone asked for 10 batches instead of 4, would counting up by adding 2 and 3 each row still be a fast way to find the answer?

SENTENCE STARTER I found the 4-batch row by multiplying ___ and ___ by ___, which is faster than adding ___ and ___ over and over ___ times.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help